

README — Direct electrochemical co-polymerization of EDOT and hydroquinone — Repository Data

Dataset Overview

This dataset contains material characterization and electrochemical experiments data supporting the study on direct electrochemical co-polymerization of EDOT and hydroquinone. The data relates to multiple figures in the manuscript and includes raw data files for both the synthesis and electrochemical characterization of PEDOT films modified with hydroquinone (QH₂) and p-toluenesulfonic acid (PTSA), as well as other characterization methods, e.g., Raman spectroscopy, X-ray photoelectron spectroscopy and FTIR.

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General Information

Data were recorded according to the procedures outlined in the manuscript.

Figure-Specific Data

Figure 1

Caption

PEDOT deposition in the presence of either QH_2 (a, c) or PTSA (b, c) additive by cyclic voltammetry (1M $LiClO_4$ in MeCN, 50 $mV s^{-1}$, 20th cycle is shown) as a CV (a, b) or a coulometric (c) response. CV response (0.1 M $HClO_4$, 50 $mV s^{-1}$) of PEDOT film (d, dashed blue curve), QH_2 solution (d, dashed violet curve), PEDOT film with PTSA additive (d, blue curve) and a co-PEDOT- QH_2 copolymer film obtained from EDOT: QH_2 2:1 mixture (d, green curve).

File Details

Filename	Description
1__HQ_cycling.csv	CV of HQ solution in 0.1 M $HClO_4$ aqueous background electrolyte, total 5 cycles in the file
1__PEDOT_cycling.csv	CV of a deposited PEDOT film in 0.1 M $HClO_4$ aqueous background electrolyte, total 5 cycles in the file
1__PTSA_17_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 17 mol.% (relative to EDOT) of PTSA in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles
1__PTSA_1_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 1 mol.% (relative to EDOT) of PTSA in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles
1__PTSA_2_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 2 mol.% (relative to EDOT) of PTSA in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles
1__PTSA_33_cycling.csv	CV of a PEDOT film in 0.1 M $HClO_4$ aqueous background electrolyte, deposited from the solution containing EDOT and 33 mol.% (relative to EDOT) of PTSA, total 5 cycles
1__PTSA_33_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 33 mol.% (relative to EDOT) of PTSA in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles
1__PTSA_5_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 5 mol.% (relative to EDOT) of PTSA in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles
1__PTSA_9_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 9 mol.% (relative to EDOT) of PTSA in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles
1__QH2_17_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 17 mol.% (relative to EDOT) of hydroquinone in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles

Filename	Description
1__QH2_1_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 1 mol.% (relative to EDOT) of hydroquinone in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles
1__QH2_26_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 26 mol.% (relative to EDOT) of hydroquinone in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles
1__QH2_2_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 2 mol.% (relative to EDOT) of hydroquinone in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles
1__QH2_33_cycling.csv	CV of a PEDOT film in 0.1 M $HClO_4$ aqueous background electrolyte, deposited from the solution containing EDOT and 33 mol.% (relative to EDOT) of hydroquinone, total 5 cycles
1__QH2_33_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 33 mol.% (relative to EDOT) of hydroquinone in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles
1__QH2_50_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 50 mol.% (relative to EDOT) of hydroquinone in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles
1__QH2_5_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 5 mol.% (relative to EDOT) of hydroquinone in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles
1__QH2_67_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 67 mol.% (relative to EDOT) of hydroquinone in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles
1__QH2_9_synth.csv	CV during the synthesis of PEDOT/PTSA film from the solution with EDOT and 9 mol.% (relative to EDOT) of hydroquinone in 1 M $LiClO_4$ acetonitrilic background solution, total 20 cycles

Figure 2

Caption

Cyclic voltammetry curves of QH_2 in MeCN (1M $LiClO_4$, u 50 $mV s^{-1}$) with variable water content (a) and with variable QH_2 concentration (b, u 500 $mV s^{-1}$)

File Details

Filename	Description
2__0%aq.csv	CV during the cycling of hydroquinone (50 $mV s^{-1}$, 1 M $LiClO_4$, AN) solution without addition of water

Filename	Description
2__0.94mM.csv	CV during the cycling of hydroquinone with the concentration of 0.94 mM (500 mV s^{-1} , 1 M LiClO ₄ , AN)
2__1%aq.csv	CV during the cycling of hydroquinone (50 mV s^{-1} , 1 M LiClO ₄ , AN) solution with addition of 1 vol.% of water
2__17%aq.csv	CV during the cycling of hydroquinone (50 mV s^{-1} , 1 M LiClO ₄ , AN) solution with addition of 17 vol.% of water
2__29%aq.csv	CV during the cycling of hydroquinone (50 mV s^{-1} , 1 M LiClO ₄ , AN) solution with addition of 29 vol.% of water
2__3%aq.csv	CV during the cycling of hydroquinone (50 mV s^{-1} , 1 M LiClO ₄ , AN) solution with addition of 3 vol.% of water
2__38%aqH+.csv	CV during the cycling of hydroquinone (50 mV s^{-1} , 1 M LiClO ₄ , AN) solution with addition of 38 vol.% of water and pH adjusted to 2 by addition of HClO_4
2__5%aq.csv	CV during the cycling of hydroquinone (50 mV s^{-1} , 1 M LiClO ₄ , AN) solution with addition of 5 vol.% of water
2__9%aq.csv	CV during the cycling of hydroquinone (50 mV s^{-1} , 1 M LiClO ₄ , AN) solution with addition of 9 vol.% of water
2__9.4mM.csv	CV during the cycling of hydroquinone with the concentration of 9.4 mM (500 mV s^{-1} , 1 M LiClO ₄ , AN)
2__940mM.csv	CV during the cycling of hydroquinone with the concentration of 940 mM (500 mV s^{-1} , 1 M LiClO ₄ , AN)
2__94mM.csv	CV during the cycling of hydroquinone with the concentration of 94 mM (500 mV s^{-1} , 1 M LiClO ₄ , AN)

Figure 3

Caption

CVs of the co-PEDOT- QH_2 films synthesized at various potentials for 200 s from 2:1 EDOT: QH_2 solution. CVs are recorded in -0.2 V to 0.9 V at 50 mV s^{-1} in 0.1 M HClO_4 . The indicated percentage values show contribution of QH_2 -related capacity towards total recorded capacity.

File Details

Filename	Description
3__0.55V.csv	Cyclic voltammetry data file
3__0.65V.csv	Cyclic voltammetry data file
3__0.75V.csv	Cyclic voltammetry data file

Filename	Description
3__0.85V.csv	Cyclic voltammetry data file
3__PEDOT_0.85V.csv	Cyclic voltammetry data file
3__PEDOT_0.75V.csv	Cyclic voltammetry data file

Figure 4

Caption

CVs (a) of co-PEDOT: QH_2 films synthesized in a potentiostatic mode (0.65 V (vs Fc/Fc^+), 200 s) from solutions with various EDOT: QH_2 ratios and the relative (b) and absolute (c) capacity contribution of QH_2 and PEDOT towards total capacity. CVs are recorded in -0.2 V to 0.9 V at 50 mV s^{-1} in 0.1 m HClO_4 .

File Details

Filename	Description
4__4.8%.csv	Cyclic voltammetry data file
4__50%.csv	Cyclic voltammetry data file
4__67%.csv	Cyclic voltammetry data file
4__83%.csv	Cyclic voltammetry data file
4__91%.csv	Cyclic voltammetry data file

Figure 5

Caption

a-c: Cyclic voltammograms and masspotentiograms for the co-PEDOT- QH_2 films pulse-deposited from 10:1 (a), 2:1 (b), and 1:2 (c) EDOT: QH_2 solutions. The stable cycle of CVs at 50 mV s^{-1} is shown. The arrows indicate direction for masspotentiograms. The currents are referred to the film mass calculated from the frequency change during the film deposition. The range for the last film is 0.1 V narrower than for the others! d-f: Mass-charge dependencies within a single CV cycle at 50 mV s^{-1} for the polymer films deposited from the solutions with the following EDOT: QH_2 ratios: (d) 10:1 ($m_{film} = 10\text{ }\mu\text{g}$), (e) 2:1 ($m_{film} = 22\text{ }\mu\text{g}$), (f) 1:2 ($m_{film} = 110\text{ }\mu\text{g}$). The arrows indicate the direction of the curve, indicated are the molar masses of pseudo-species transferred in the highlighted range.

File Details

Filename	Description
5__9%.csv	Cyclic voltammetry with operando EQCM data file
5__33%.csv	Cyclic voltammetry with operando EQCM data file
5__67%.csv	Cyclic voltammetry with operando EQCM data file

Figure 6

Caption

Galvanostatic charge-discharge curves recorded for co-PEDOT: QH_2 films synthesized from 1:2 EDOT: QH_2 solution in pulse deposition mode on a quartz crystal (a), and discharge specific capacity values for the same film at different current densities (b), capacity retention and coulombic efficiency of co-PEDOT- QH_2 films synthesized from solutions with different EDOT: QH_2 ratios (c).

File Details

Filename	Description
6__0.1Ag.csv	Galvanostatic charge-discharge with operando EQCM data file
6__0.2Ag.csv	Galvanostatic charge-discharge with operando EQCM data file
6__0.5Ag.csv	Galvanostatic charge-discharge with operando EQCM data file
6__1Ag.csv	Galvanostatic charge-discharge with operando EQCM data file
6__2Ag.csv	Galvanostatic charge-discharge with operando EQCM data file
6__5Ag.csv	Galvanostatic charge-discharge with operando EQCM data file

Figure 7

Caption

Normalized cyclic voltammograms of co-PEDOT: QH_2 film synthesized from the solution with 1:2 EDOT: QH_2 ratio at different scan rates indicated in a color bar ($V\ s^{-1}$) (a), bilogarithmic dependence of peak currents on scan rate to determine capacitive and diffusion currents contribution (b).highlighted range.

File Details

Filename	Description
7__0.00928.csv	Cyclic voltammetry data

Filename	Description
7__0.0120.csv	Cyclic voltammetry data
7__0.0155.csv	Cyclic voltammetry data
7__0.02.csv	Cyclic voltammetry data
7__0.0258.csv	Cyclic voltammetry data
7__0.0334.csv	Cyclic voltammetry data
7__0.0431.csv	Cyclic voltammetry data
7__0.0557.csv	Cyclic voltammetry data
7__0.0719.csv	Cyclic voltammetry data
7__0.0928.csv	Cyclic voltammetry data
7__0.120.csv	Cyclic voltammetry data
7__0.155.csv	Cyclic voltammetry data
7__0.2.csv	Cyclic voltammetry data
7__0.258.csv	Cyclic voltammetry data
7__0.334.csv	Cyclic voltammetry data
7__0.431.csv	Cyclic voltammetry data
7__0.557.csv	Cyclic voltammetry data
7__0.719.csv	Cyclic voltammetry data
7__0.928.csv	Cyclic voltammetry data
7__1.20.csv	Cyclic voltammetry data
7__1.55.csv	Cyclic voltammetry data
7__11.99.csv	Cyclic voltammetry data
7__15.49.csv	Cyclic voltammetry data
7__2.58.csv	Cyclic voltammetry data
7__2.csv	Cyclic voltammetry data
7__20.csv	Cyclic voltammetry data
7__3.34.csv	Cyclic voltammetry data
7__4.31.csv	Cyclic voltammetry data
7__5.57.csv	Cyclic voltammetry data
7__7.19.csv	Cyclic voltammetry data
7__9.28.csv	Cyclic voltammetry data

Figure 8

Caption

Sulfur atomic percentage according to XPS data vs. PEDOT contribution according to cyclic voltammetry data. XPS data are calibrated by the theoretical sulfur atomic percentage expected for PEDOT polymer.

File Details

Filename	Description
8__survey_data.csv	Data compiled from XPS and CV experiments

Figure 9

Caption

2D colormaps of UV-Vis spectra for the samples of PEDOT and co-PEDOT:QH₂ films synthesized from the solutions with various EDOT:QH₂ ratios. The colormaps visualize the oxidation and reduction processes within a single CV cycle at 50 mV s⁻¹ from -0.2 V to 0.9 V and back to -0.2 V.

File Details

Filename	Description
9__33%QH2_lambda-vs-intensity-at-potential.csv	3D (absorbance vs wavelength vs potential) spectroelectrochemical data
9__36%QH2_lambda-vs-intensity-at-potential.csv	3D (absorbance vs wavelength vs potential) spectroelectrochemical data
9__40%QH2_lambda-vs-intensity-at-potential.csv	3D (absorbance vs wavelength vs potential) spectroelectrochemical data
9__50%QH2_lambda-vs-intensity-at-potential.csv	3D (absorbance vs wavelength vs potential) spectroelectrochemical data
9__67%QH2_lambda-vs-intensity-at-potential.csv	3D (absorbance vs wavelength vs potential) spectroelectrochemical data
9__PEDOT_lambda-vs-intensity-at-potential.csv	3D (absorbance vs wavelength vs potential) spectroelectrochemical data

Figure 10

Caption

Conductance curves recorded in 0.1 M HClO₄ at 50 mV s⁻¹) for co-PEDOT-QH₂ films synthesized from 1:2 (a), 2:1 (b) and 10:1 (c) EDOT:QH₂ solution in pulse deposition mode on IDE chips.

File Details

Filename	Description
10_33%QH2_2000p_IDE.csv	Cyclic voltammetry data recorded on a bipotentiostat for an IDE electrode coated by the polymer film
10_67%QH2_1000p_IDE.csv	Cyclic voltammetry data recorded on a bipotentiostat for an IDE electrode coated by the polymer film
10_67%QH2_2000p_IDE.csv	Cyclic voltammetry data recorded on a bipotentiostat for an IDE electrode coated by the polymer film
10_67%QH2_500p_IDE.csv	Cyclic voltammetry data recorded on a bipotentiostat for an IDE electrode coated by the polymer film
10_9%QH2_2000p_IDE.csv	Cyclic voltammetry data recorded on a bipotentiostat for an IDE electrode coated by the polymer film

Figure 11

Caption

EIS spectra recorded 0.1 M $HClO_4$ for co-PEDOT- QH_2 films synthesized from 1:2 and 10:1 EDOT: QH_2 solutions in pulse deposition mode shown in mid-to-low frequencies (a) and high-to-mid frequencies (b). The equivalent scheme used for the fitting is shown above the spectra. The R_s is subtracted from the real part of the spectra for visual clarity.

File Details

Filename	Description
11_67%QH2_FRA.csv	FRA data (frequency, real, and imaginary components of impedance) recorded at 0.025 V vs Ag pseudo-reference electrode
11_9%QH2_FRA.csv	FRA data (frequency, real, and imaginary components of impedance) recorded at 0.025 V vs Ag pseudo-reference electrode

File Naming Conventions

Figure 1

Files are named according to the following pattern:

`1__[additive]_[concentration]_[type].csv`

Where:

- `[additive]` indicates the type of additive (QH_2 for hydroquinone, PTSA for p-toluenesulfonic acid)

- `[concentration]` indicates the molar percentage of additive relative to EDOT
- `[type]` indicates whether the file contains synthesis data ("synth") or characterization data ("cycling")

Figure 2

Files are named according to the following pattern:

`2__[percentage-added]%aq_{H+}.csv`

or

`2__[molarity]mM.csv`

Where:

- `[percentage-added]` indicates the amount of water added in volumetric % to the cycling solution
- `{H+}` optional indicates whether the solution was acidified to the pH of 2
- `[molarity]` indicates the molar concentration of hydroquinone in the solution

Figure 3

Files are named according to the following pattern:

`3__[deposition-potential].csv`

Where:

- `[deposition-potential]` indicates the potentiostatic deposition potential of the film vs Fc/Fc^+ redox pair.

Figure 4

Files are named according to the following pattern:

`4__[QH2-percentage]%.csv`

Where:

- `[QH2-percentage]` indicates the molar percentage of QH_2 in the synthesis solution used for electrochemical polymerization of the material

Figure 5

Files are named according to the following pattern:

`5__[QH2-percentage]%QH2.csv`

Where:

- `[QH2-percentage]` indicates the molar percentage of QH_2 in the synthesis solution used for electrochemical polymerization of the material

Figure 6

Files are named according to the following pattern:

`6__[current-density]Ag.csv`

Where:

- `[current-density]` indicates the current density in (A/g) applied during the GCD procedure performed.

Figure 7

Files are named according to the following pattern:

`7__[scan-rate].csv`

Where:

- `[scan-rate]` indicates the scan rate (in $V\ s^{-1}$) used to record the CV in the given file.

Figure 8

`8__survey_XPS_vs_CVs.csv`

A single file contains all the data plotted in the figure.

Figure 9

`9__[percentage]QH2_lambda-vs-intensity-at-potential.csv`

Where:

- `[percentage]` indicates the molar percentage of QH_2 in the synthesis solution used for electrochemical polymerization of the material, with 00% related to pristine PEDOT film

Figure 10

`10__[percentage]QH2_[pulse-number]p_IDE.csv`

Where:

- `[percentage]` indicates the molar percentage of QH_2 in the synthesis solution used for electrochemical polymerization of the copolymer
- `[pulse-number]` indicates the number of pulses used in the pulse potentiostatic deposition of the copolymer

Figure 11

`11__[percentage]QH2_FRA.csv`

Where:

- `[percentage]` indicates the molar percentage of QH_2 in the synthesis solution used for electrochemical polymerization of the copolymer

Data Format

For all .csv files, the columns are comma-separated.

For most files, the first row of the file contains the quantity or parameter name, and the second row contains the units, if applicable.

For Figure 2:

- %aq files: There is a single cycle presented recorded at 50 mV s^{-1} , and the columns are potential and current.

For Figures 5, 6:

- The third row contains additionally the conversion factor from the raw data to actual measured quantity, if applicable.

For Figure 8:

- The file contains 9 columns titled EDOTmol% S2p PEDOT% PEDOT charge Full charge S2p EDOTmol% S2p PEDOT%

For Figure 9:

- The file contains an array where the header row indicates the potential (V) and the first column indicates the wavelength (nm) at which the the absorbance intensity (rel.u., background-subtracted by the internal mechanism of the hardware (background is pristine ITO-coated glass submerged in a quartz cuvette filled with the electrolyte)) was recorded.

For Figure 10:

- The file contains the potential-current pairs for both working electrodes of the IDE recording on a bipotentiostat simultaneously.

For Figure 11:

- The file contains the real part (Ω), negative imaginary part of impedance (Ω), and frequency (Hz).

Supplementary Information

The data for the Supplementary Information materials is partially provided for the following methods:

- **raman** — Raman spectroscopy data for the samples of the films synthesized from the solutions containing different concentrations of QH_2 , as per experimental description in the text. The first and the second columns of each file contain Raman shift (cm^{-1}) and intensity (arbitrary units), respectively.
- **ftir** — FT-IR data (ATR and KBr) for the samples of the films synthesized from the solutions containing different concentrations of QH_2 , as per experimental description in the text. The first and the second columns of each file contain wave number (cm^{-1}) and transmittance (%), respectively.
- **ms** — mass spectrometry data for co-PEDOT- QH_2 copolymer converted to a standard format (**.mzML**) from Bruker (**.d**) format using **MSConvert** by **Proteowizard**.

Equipment and Software

Most electrochemical data were recorded using Autolab PGSTAT128N. For conductance measurements, DropSens μ 400 bipotentiostat was used. UV-Vis spectra were recorded using Avantes AvaSpec.

The data were prepared using proprietary software available with the equipment, and then converted to common file types using Python (>3.10) and OriginPro (>9.0).

Date of Data Collection

The data were collected over a period spanning from February 2023 to April 2025.

Contact Information

For questions regarding this dataset, please contact: a.i.volkov@spbu.ru

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